

FRESHMAN SEMINAR · ANSWER KEY - PROFESSOR COPY

Homework 2: Big-Idea Hunting

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Boil it down

From NICU Tutor: Drinking milk seems like the easiest thing a baby does. It's actually a stunt. To feed safely, a baby must suck, then swallow, then breathe — in perfect rhythm, over and over, without ever letting milk slip down the wrong pipe into the lungs. That rhythm is a brain skill, and it isn't wired up until about 32 to 34 weeks of development. A baby born at 26 weeks simply doesn't have the wiring yet. Feed that baby from a bottle and milk would end up in its lungs. So the NICU delivers milk another way: a soft tube from the nose to the stomach, with a pump pushing the milk in slowly.

1. Write the ONE biggest idea of that passage in your own words — one line. Then, underneath, write one detail you chose to leave out. (3 pts)

Answer: Idea (2 pts): feeding by mouth needs a suck-swallow-breathe rhythm that early preemies' brains haven't wired yet, so they're fed by a tube instead. Detail (1 pt): any support — '32 to 34 weeks', 'the wrong pipe', 'a pump pushing slowly', etc. Give 1 pt only if the 'idea' is actually a detail (like the 32-34 weeks number).

From this morning's Chemistry: Look back at the ruler. Blood sits at 7.35 to 7.45. That window is one-tenth of one step wide. Your stomach is happy at 2; your skin is fine at 5; but blood that drifts to 7.2 makes a NICU team move fast, and blood at 6.9 or 7.8 is usually fatal. Why so picky? Because of proteins. A protein works because of its exact folded shape, and loose hydrogens grab onto proteins and warp them. A little too much acid and the enzymes slow down, the heart muscle squeezes more weakly, and hemoglobin drops its oxygen too early.

2. Same again: the ONE big idea, one line, your words. Then one detail you left out. (3 pts)

Answer: Idea (2 pts): blood must stay in a tiny pH window (about 7.4) because too much acid warps the proteins the body runs on. Detail (1 pt): 'stomach at 2', '6.9 or 7.8 is fatal', 'hemoglobin drops oxygen early', etc.

Part B — Idea or detail?

3. Six sentences from this week's Monday classes. Mark each one I (big idea) or D (detail), and say which test you used (so what? / forget it / repeat). (6 pts)

Answer: D, I, I, D, I, D. 1 pt per row for the right letter; any sensible test named is fine (the 'forget it' test works on all six).

Part C — You're the professor

4. Write ONE multiple-choice quiz question about something from today's Chemistry or Biology. Include the question, the right answer, and TWO wrong answers that sound reasonable (no jokes, and about the same length as the right one). Remember the rule: someone should have to actually know the material. (4 pts)

Answer: 1 pt: the question is about a big idea, not a trivial detail. 1 pt: the right answer is actually right. 2 pts: both wrong answers are plausible and similar in length (a kid who half-understood might pick them). Take a point off for a joke answer or one that's obviously shorter/sillier.

Part D — Flashbacks

5. From NICU Tutor: what does the isolette's skin probe do? (1 pt)

Answer: Reports the baby's skin temperature so the isolette can warm or cool its air to hit the target (servo control).

6. From last week's Seminar: what are office hours? (1 pt)

Answer: Time a professor sets aside for students to come and ask anything.

Part E — Think like a student

7. A friend says: "I reread the chapter three times and still failed the quiz." Using today's class, tell them what to do instead — and why it works better. (3 pts)

Answer: Close the book and test yourself: say the big ideas out loud, write them from memory, teach them to someone, or write your own quiz questions (1 pt). Why: pulling ideas out of your head (retrieval) builds the memory path; rereading only makes words look familiar (1 pt). Bonus for mentioning the 2006 study (about 60% vs 40%) or 'find the 3 big ideas first, then test those' (1 pt).